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Control device and control method

Abstract

A control device controls a robot and an articulated shaft of the robot, the articulated shaft being attached to the distal end of an arm of the robot. The control device includes an operation program analysis unit that calculates a movable range of a combination of the robot and articulated shaft based on an operation program, and a robot/tool moving distance calculation unit that, based on the calculated movable range, sets an amount of movement of the articulated shaft for moving the articulated shaft to a boundary position as a specified value, and calculates the amount of movement of the distal end of the arm when the articulated shaft is moved to the boundary position, and the amounts of movement of the articulated shaft and the distal end of the arm of the robot for at least one teaching point other than the boundary position.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a control device and a control method.

BACKGROUND ART

(2) A technique of conveying a workpiece at high speed has been known. In this technique, a sliding articulated shaft (may be hereinafter referred to as a “linear tool”) that moves the workpiece linearly in one direction (e.g., in a Y-axis direction) is attached to an end of an arm of a six-axis robot to modify the robot into a seven-axis robot, and the robot and the linear tool are moved in the same direction. For example, see Patent Document 1.

(3) To teach a seven-axis robot system, in general, travel distances of the six-axis robot and the linear tool are specified separately. Patent Document 1: Japanese Unexamined Patent Application, Publication No. 2019-126850

DISCLOSURE OF THE INVENTION

Problems to be Solved by the Invention

(4) The six-axis robot equipped with the linear tool is generally taught its positions and postures at teaching points while the robot is moving.

(5) FIG. 8 is a view illustrating an example of the seven-axis robot system having the linear tool attached to the end of the arm of the six-axis robot.

(6) As shown in FIG. 8, a robot **10** is, for example, a six-axis vertically articulated robot, and has a linear tool **20** at an end of its arm. The linear tool **20** is, for example, a conveying tool in the shape of a sliding arm (an articulated shaft), and has a strip-shaped (a rectangular plate-shaped) frame **21**, a robot-side slider **22** arranged on an upper surface of the frame **21**, and a workpiece-side slider **23** arranged on a lower surface of the frame **21**. A tool (not shown) having a plurality of adsorption pads that adsorb the workpiece, for example, is attached to the workpiece-side slider **23**, and the linear tool **20** is able to linearly move the workpiece (not shown) adsorbed by the attached tool (not shown) in one direction, i.e., a Y-axis direction.

(7) FIG. 9 is a view schematically illustrating an example of positional relationship between the end of the arm of the robot **10** and the linear tool **20**. FIG. 9 shows, as the robot **10**, the end of the arm of the robot **10** (may be hereinafter referred to as a “finger of the robot **10**”) and other parts of the robot **10** than the finger. In FIG. 9, a coordinate origin of the robot **10**, the position of the finger of the robot **10**, a coordinate origin of the tool, and the position of the workpiece-side slider **23** of the linear tool **20** are on a straight line in an X-axis direction.

(8) FIG. 10 is a view schematically illustrating an example of a positional relationship when the workpiece-side slider **23** of the linear tool **20** moves in a +Y-axis direction.

(9) As shown in FIG. 10, when the workpiece-side slider **23** of the linear tool **20** moves, for example, in the +Y-axis direction, the finger of the robot **10** moves in the +Y-axis direction by a distance $D_{sub.r}$, which is a finger's travel amount, and the workpiece-side slider **23** moves in the +Y-axis direction by a distance D_t , which is a tool's travel amount relative to the coordinate origin of the tool. To create operation programs of the robot **10**, an operator needs to examine the finger's travel amount $D_{sub.r}$ and the tool's travel amount $D_{sub.t}$ and input a total travel amount $D_{sub.all}$ ($=D_{sub.r}+D_{sub.t}$) representing the coordinates of the workpiece-side slider **23** relative to the coordinate origin of the robot **10** (may be hereinafter referred to as “robot and tool's coordinates”) and the tool's travel amount D_t .

(10) However, teaching experience is required for the examination of the finger's travel amount $D_{sub.r}$ and the tool's travel amount $D_{sub.t}$. If the operator has little experience, the teaching disadvantageously requires more man-hours and longer time.

(11) Under these circumstances, it has been required that the operation programs are created by teaching a single teaching point only, irrespective of the experience of the operator.

Means for Solving the Problems

(12) A control device according to an aspect of the present disclosure controls a robot and an

articulated shaft that is attached to an end of an arm of the robot and moves a workpiece in one direction, the control device comprising: an operation program analysis unit configured to calculate a total travel range of the robot and the articulated shaft based on an operation program that sets an initial position of the articulated shaft relative to a coordinate origin of the robot; and a robot/tool travel distance calculation unit configured to set a travel amount that the articulated shaft would be required to take to reach a boundary position of the travel range to a specified value based on the calculated travel range, and calculate a travel amount of the end of the arm of the robot when the articulated shaft has traveled to the boundary position and travel amounts that the articulated shaft and the end of the arm of the robot would be required to take to reach at least one teaching point other than the boundary position.

(13) A control method according to an aspect of the present disclosure controls a robot and an articulated shaft that is attached to an end of an arm of the robot and moves a workpiece in one direction, the control method comprising: an operation program analysis step of calculating a total travel range of the robot and the articulated shaft based on an operation program that sets an initial position of the articulated shaft relative to a coordinate origin of the robot; and a robot/tool travel distance calculation step of setting a travel amount that the articulated shaft would be required to take to reach a boundary position of the travel range to a specified value based on the calculated travel range, and calculating a travel amount of the end of the arm of the robot when the articulated shaft has traveled to the boundary position and travel amounts that the articulated shaft and the end of the arm of the robot would be required to take to reach at least one teaching point other than the boundary position.

Effects of the Invention

(14) According to the aspect, operation programs can be created by teaching a robot a single teaching point only, irrespective of the experience of an operator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a functional block diagram illustrating an example of functional configuration of a robot system according to an embodiment;
- (2) FIG. 2 is a view illustrating an example of an operation program;
- (3) FIG. 3 is a view illustrating an example of a teaching point set in the operation program of FIG. 2;
- (4) FIG. 4A is a view illustrating an example of positional relationship between a robot finger and a workpiece-side slider of a linear tool at teaching points P1 and P6;
- (5) FIG. 4B is a view illustrating an example of positional relationship between the robot finger and the workpiece-side slider of the linear tool at a teaching point P3;
- (6) FIG. 5 is a view illustrating an example of relationship between the teaching points and a coordinate origin of the robot;
- (7) FIG. 6 is a flowchart illustrating control processing by a robot control device;
- (8) FIG. 7 is a view illustrating an example of relationship between the teaching points and boundary positions;
- (9) FIG. 8 is a view illustrating an example of a seven-axis robot system having a linear tool attached to an end of an arm of a six-axis robot;
- (10) FIG. 9 is a view schematically illustrating an example of positional relationship between the end of the arm of the robot and the linear tool; and
- (11) FIG. 10 is a view schematically illustrating an example of the positional relationship when the workpiece-side slider of the linear tool moves in a +Y-axis direction.

PREFERRED MODE FOR CARRYING OUT THE INVENTION

(12) An embodiment will be described below with reference to the drawings.

(13) <Embodiment>

(14) FIG. 1 is a functional block diagram illustrating an example of functional configuration of a robot system according to an embodiment. Components having the same functions as the robot **10** and the components of the linear tool **20** shown in FIG. 8 are designated by the same reference numerals, and will not be described in detail below.

(15) As shown in FIG. 1, a robot system **1** includes a robot **10**, a linear tool **20**, and a robot control device **30**. For example, the robot system **1** is used in a factory having many press machines (not shown), and the robot **10** conveys a workpiece between the pressing machines which are not shown using the linear tool **20** based on a control command from the robot control device **30**.

(16) The robot **10** and the linear tool **20** may be directly connected to the robot control device **30** via a connection interface which is not shown. The robot **10** and the linear tool **20** may be mutually connected to the robot control device **30** via a network such as a local area network (LAN). In this case, a communication unit which is not shown may be provided for the robot **10** and the linear tool **20**, and the robot control device **30** as well, to achieve mutual communication via the connection.

(17) <Robot Control Device **30**>

(18) As shown in FIG. 1, the robot control device **30** outputs a drive command to the robot **10** and the linear tool **20** based on an operation program to control the motions of the robot **10** and the linear tool **20**. The robot control device **30** may be connected to a teaching console (not shown) which teaches the motions to the robot **10** and the linear tool **20**.

(19) As shown in FIG. 1, the robot control device **30** of the present embodiment includes a control unit **31**, an input unit **32**, a storage unit **33**, and a display unit **34**. The control unit **31** includes an operation program analysis unit **311**, a robot/tool travel distance calculation unit **312**, and an operation control/drive unit **313**.

(20) The input unit **32** is, for example, a keyboard or a button which is not shown and included in the robot control device **30** or a touch screen of the display unit **34** which will be described later, and receives an input from an operator of the robot control device **30**.

(21) The storage unit **33** is, for example, a read only memory (ROM) or a hard disk drive (HDD), and stores programs such as a system program and an application program executed by the control unit **31** which will be described later. The storage unit **33** may store a finger's travel amount $D_{sub.r}$, which is a travel amount of the finger of the robot **10**, and a tool's travel amount $D_{sub.t}$, which is a travel amount of the workpiece-side slider **23** of the linear tool **20**, to teaching points calculated by the robot/tool travel distance calculation unit **312** which will be described later.

(22) The display unit **34** is, for example, a display device such as a liquid crystal display (LCD), and shows a screen such as a screen on which the operation program is created and a screen on which the tool's travel amount $D_{sub.t}$ of the workpiece-side slider **23** is specified.

(23) <Control Unit **31**>

(24) The control unit **31** includes components such as a central processing unit (CPU), a ROM, a RAM, and a complementary metal-oxide-semiconductor (CMOS) memory, which are known in the art and are able to communicate with each other via buses. The CPU is a processor that generally controls the robot control device **30**.

(25) The CPU reads the system program and the application program stored in the ROM via the bus, and controls the whole robot control device **30** in accordance with the system program and the application program. Thus, as shown in FIG. 1, the control unit **31** is configured to achieve the functions of the operation program analysis unit **311**, the robot/tool travel distance calculation unit **312**, and the operation control/drive unit **313**. The RAM stores various types of data such as temporary calculation data and display data. The CMOS memory is backed up by a battery which is not shown, and is configured as a nonvolatile memory that stores the data after the power of the robot control device **30** is turned off.

(26) The operation program analysis unit **311** receives, for example, an input of an operation

program that sets the coordinates of the workpiece-side slider **23** relative to the coordinate origin of the robot **10** (i.e., the robot and tool's coordinates) from an external device which is not shown, such as a computer, and calculates a total travel range of the robot **10** and the linear tool **20** based on the inputted operation program.

(27) The operation program analysis unit **311** may acquire an operation program created by the input from the operator via the input unit **32**.

(28) FIG. **2** is a view illustrating an example of the operation program. The operation program of FIG. **2** sets, as described above, the robot and tool's coordinates (e.g., the maximum distance that the workpiece-side slider **23** can travel relative to the coordinate origin of the robot **10** or the value specified by the operator) only in "position [1]" corresponding to the teaching point **P1**.

(29) FIG. **3** is a view illustrating an example of the teaching points set in the operation program of FIG. **2**. Specifically, based on the operation program shown in FIG. **2**, the robot **10** and the linear tool **20** move from, for example, the teaching point **P1** indicating the robot and tool's coordinates to the teaching points **P2** to **P6** in this order. FIG. **2** only shows "position [1]" and "position [2]" corresponding to the teaching points **P1** and **P2** of the operation program, but the operation program also includes "position [3]" to "position [6]" corresponding the teaching points **P3** to **P6**.

(30) FIG. **4A** is a view illustrating an example of positional relationship between the finger of the robot **10** and the workpiece-side slider **23** of the linear tool **20** at the teaching points **P1** and **P6**.

FIG. **4B** is a view illustrating an example of positional relationship between the finger of the robot **10** and the workpiece-side slider **23** of the linear tool **20** at the teaching point **P3**. The tool's travel amounts of the workpiece-side slider **23** from the coordinate origin of the tool (the position of the finger of the robot **10**) to the teaching points **P1**, **P3**, and **P6** will be referred to as distances $D_{sub.t1}$, $D_{sub.t3}$, and $D_{sub.t6}$.

(31) The operation program shown in FIG. **2** includes six teaching points, but the number of teaching points may be other than six.

(32) Specifically, the operation program analysis unit **311** analyzes the inputted operation program. Based on the robot and tool's coordinates at the teaching point **P1** set in the "position [1]" of the operation program, the operation program analysis unit **311** calculates a travel range from one boundary position that is the teaching point **P1** to the other boundary position that is the teaching point **P6** as shown in FIG. **4A** in an area surrounded by the teaching points **P1** to **P6** shown in FIG. **3**. The one boundary position is the farthest position of the linear tool **20** in the $-Y$ -axis direction, and the other boundary position is the farthest position of the linear tool **20** in the $+Y$ -axis direction.

(33) The robot/tool travel distance calculation unit **312** sets the tool's travel amount that the workpiece-side slider **23** of the linear tool **20** takes to reach each boundary position of the travel range to a specified value based on the calculated travel range, and calculates a travel amount of the finger of the robot when the workpiece-side slider **23** has traveled to each boundary position and a travel amount of the linear tool **20** to each of the teaching points **P2** to **P5** other than the boundary positions.

(34) Specifically, for example, the robot/tool travel distance calculation unit **312** regards a distance $D_{sub.t1}$ from the coordinate origin of the tool to the coordinates of the workpiece-side slider **23** at the teaching point **P1** which is the boundary position and a distance $D_{sub.t6}$ from the coordinate origin of the tool to the coordinates of the workpiece-side slider **23** at the teaching point **P6** which is also the boundary position as the tool's travel amounts, and sets the tool's travel amount $D_{sub.t1}$ to a specified value $D_{sub.c1}$, and the tool's travel amount $D_{sub.t6}$ to a specified value $D_{sub.c2}$. The robot/tool travel distance calculation unit **312** may calculate the travel amount of the finger of the robot **10** to the teaching point **P1** which is the boundary position from the difference between the robot and tool's coordinates at the teaching point **P1** and the specified value $D_{sub.c1}$, and calculate the travel amount of the finger of the robot **10** to the teaching point **P6** which is the boundary position from the difference between the robot and tool's coordinates at the teaching point **P6** and the specified value $D_{sub.c2}$.

- (35) Each of the specified values D.sub.c1 and D.sub.c2 may be the maximum travel amount of the workpiece-side slider **23** relative to the coordinate origin of the tool, or may be a travel amount specified by the operator. For example, when the specified values D.sub.c1 and D.sub.c2 are the maximum travel amounts of the workpiece-side slider **23** relative to the coordinate origin of the tool, the performance of the linear tool **20** can be maximized.
- (36) Next, the robot/tool travel distance calculation unit **312** calculates the tool's travel amount D.sub.tn to the teaching points Pn (n=2, 3, 4, and 5) other than the boundary positions **P1** and **P6** using the formula 1 or 2.
- (37) $D_{tn} = D_{C1} \times \frac{D_{yn}}{D_{max1}}$ [Formula1] $D_{tn} = D_{C2} \times \frac{D_{yn}}{D_{max2}}$ [Formula2]
- (38) In the formulae, D.sub.max1 represents Y-axis coordinates of the teaching point **P1** which is the boundary position relative to the coordinate origin of the robot **10**, and D.sub.max2 represents Y-axis coordinates of the teaching point **P6** which is the boundary position relative to the coordinate origin of the robot **10**. Specifically, D.sub.max1 is a robot and tool's Y-axis travel amount D.sub.Y1 which is the maximum travel amount in the -Y-axis direction, and D.sub.max2 is a robot and tool's Y-axis travel amount D.sub.Y6 which is the maximum travel amount in the +Y-axis direction. D.sub.yn represents the robot and tool's coordinates at the teaching point Pn relative to the coordinate origin of the robot **10** (may be hereinafter referred to as a "Y-axis travel amount").
- (39) Specifically, when the robot and tool's Y-axis travel amount D.sub.yn to the teaching point Pn is equal to or less than "0" as shown in FIG. 5, the robot/tool travel distance calculation unit **312** calculates the tool's travel amount D.sub.tn to the teaching point Pn from the ratio between the Y-axis travel amount D.sub.max1 to the teaching point **P1** and the Y-axis travel amount D.sub.yn to the teaching point Pn using the formula 1. For example, when the Y-axis travel amount D.sub.max1 to the teaching point **P1** relative to the coordinate origin of the robot **10** is 1000 mm and the tool's travel amount Dt1 (specified value D.sub.c1) is 800 mm, the tool's travel amount D.sub.t3 is 400 mm when the Y-axis travel amount D.sub.y3 to the teaching point **P3** is 500 mm.
- (40) When the robot and tool's Y-axis travel amount D.sub.yn to the teaching point Pn is greater than "0", the robot/tool travel distance calculation unit **312** calculates the tool's travel amount D.sub.tn to the teaching point Pn from the ratio between the Y-axis travel amount D.sub.max2 to the teaching point **P6** and the Y-axis travel amount D.sub.yn to the teaching point Pn using the formula 2.
- (41) The robot/tool travel distance calculation unit **312** may calculate the travel amount of the finger of the robot **10** to the teaching point Pn other than the boundary positions from the difference between the robot and tool's coordinates at the teaching point Pn other than the boundary positions and the tool's travel amount D.sub.tn.
- (42) The operation control/drive unit **313** sets the travel amounts of the finger of the robot **10** and the tool's travel amounts D.sub.tn to the teaching points **P2** to **P6** calculated by the robot/tool travel distance calculation unit **312** in "position [2]" to "position [6]" of the operation program, and executes the operation program to drive the robot **10** and the linear tool **20**.
- (43) <Control Processing by Robot Control Device **30**>
- (44) Control processing by the robot control device **30** of the present embodiment will be described below.
- (45) FIG. 6 is a flowchart illustrating the control processing by the robot control device **30**. The processing shown in the flowchart is performed every time the operation program that sets the coordinates of the workpiece-side slider **23** (the robot and tool's coordinates) relative to the coordinate origin of the robot **10** at the teaching point **P1** is inputted.
- (46) In Step **S1**, the operation program analysis unit **311** receives an input of the operation program that sets the robot and tool's coordinates at the teaching point **P1** from an external device (not shown).
- (47) In Step **S2**, the operation program analysis unit **311** analyzes the inputted operation program

and calculates a travel range having the teaching points **P1** and **P6** as the boundary positions based on the robot and tool's coordinates at the teaching point **P1** set in “position [1]” of the operation program.

(48) In Step **S3**, the robot/tool travel distance calculation unit **312** sets the tool's travel amounts $D_{sub.t1}$ and $D_{sub.t6}$ of the workpiece-side slider **23** to the teaching points **P1** and **P6** to the specified values $D_{sub.c1}$ and $D_{sub.c2}$.

(49) In Step **S4**, the robot/tool travel distance calculation unit **312** calculates the travel amount of the finger of the robot **10** to the teaching point **P1** which is the boundary position from the difference between the robot and tool's coordinates at the teaching point **P1** (Y-axis travel amount $D_{sub.max1}$) and the specified value $D_{sub.c1}$, and calculates the travel amount of the finger of the robot **10** to the teaching point **P6** which is the boundary position from the difference between the robot and tool's coordinates at the teaching point **P6** (Y-axis travel amount $D_{sub.max2}$) and the specified value $D_{sub.c2}$.

(50) In Step **S5**, the robot/tool travel distance calculation unit **312** calculates the tool's travel amounts $D_{sub.tn}$ to the teaching points P_n other than the boundary positions **P1** and **P6** using the formula 1 or 2.

(51) In Step **S6**, the robot/tool travel distance calculation unit **312** calculates the travel amount of the finger of the robot **10** to each of the teaching points P_n other than the boundary positions **P1** and **P6** from the difference between the robot and tool's coordinates (Y-axis travel amount $D_{sub.yn}$) at the teaching point P_n other than the boundary positions **P1** and **P6** and the corresponding tool's travel amount $D_{sub.tn}$.

(52) In Step **S7**, the operation control/drive unit **313** sets the calculated travel amounts of the finger of the robot **10** to the teaching points **P2** to **P6** and the calculated tool's travel amounts $D_{sub.tn}$ in “position [2]” to “position [6]” of the operation program, and executes the operation program to drive the robot **10** and the linear tool **20**.

(53) As described above, the robot control device **30** of the embodiment receives an input of the operation program that sets the robot and tool's coordinates at the teaching point **P1**, which is one of the farthest points in the travel range of the workpiece-side slider **23** of the linear tool **20**. Based on the robot and tool's coordinates at the teaching point **P1**, the robot control device **30** calculates the travel amounts of the finger of the robot **10** and the tool's travel amounts to the other teaching points **P2** to **P6** using the formula 1 or 2, and sets the calculated travel amounts in the operation program. With this robot control device **30**, an operator can create the operation program by teaching the single teaching point only, irrespective of the experience of the operator. This allows an operator with little experience in the teaching of the robot and the tool to achieve the teaching easily, saving time and effort of the operator to teach the position of the tool.

(54) The embodiment has just been described above, but the robot control device **30** is not limited to the embodiment described above, and may be modified or improved to the extent that the purpose can be achieved.

(55) <First Variation>

(56) It has been described in the embodiment that the linear tool **20** moves in the Y-axis direction. However, the moving direction is not limited to the Y-axis direction, and the linear tool **20** may move in any direction in an X-Y plane.

(57) <Second Variation>

(58) Although it has been described in the embodiment that the robot/tool travel distance calculation unit **312** calculates the tool's travel amounts $D_{sub.tn}$ to the teaching points P_n other than the boundary positions **P1** and **P6** using the formula 1 or 2. However, the present invention is not limited to this example. For example, the robot/tool travel distance calculation unit **312** may calculate the tool's travel amounts $D_{sub.tn}$ to the teaching points P_n other than the boundary positions **P1** and **P6** using the formula 3 or 4. Specifically, when the Y-axis travel amount $D_{sub.yn}$ to the teaching point P_n is equal to or less than $(D_{sub.max1} + D_{sub.max2})/2$ as shown in FIG. 7,

the robot/tool travel distance calculation unit **312** calculates the tool's travel amount $D_{\text{sub.tn}}$ to the teaching point P_n using the formula 3. Alternatively, when the Y-axis travel amount $D_{\text{sub.yn}}$ to the teaching point P_n is more than $(D_{\text{sub.max1}} + D_{\text{sub.max2}})/2$, the robot/tool travel distance calculation unit **312** calculates the tool's travel amount $D_{\text{sub.tn}}$ to the teaching point P_n using the formula 4.

$$(59) \quad D_{\text{tn}} = D_{C1} \times \frac{2D_{\text{yn}} - D_{\text{max1}} - D_{\text{max2}}}{D_{\text{max1}} - D_{\text{max2}}} \quad [\text{Formula3}] \quad D_{\text{tn}} = D_{C2} \times \frac{2D_{\text{yn}} - D_{\text{max1}} - D_{\text{max2}}}{D_{\text{max2}} - D_{\text{max1}}} \quad [\text{Formula4}]$$

(60) The functions of the robot control device **30** of the embodiment can be implemented by hardware, software, or a combination of the hardware and the software. The expression “the functions are implemented by the software” means that a computer reads and executes a program to implement the functions.

(61) The components of the robot control device **30** can be implemented by hardware including an electronic circuit, software, or a combination of the hardware and the software.

(62) The program is stored in various types of non-transitory computer readable media and supplied to the computer. The non-transitory computer readable media include various types of tangible storage media. Examples of the non-transitory computer readable media include a magnetic recording medium (e.g., a flexible disk, a magnetic tape, and a hard disk drive), a magneto-optical recording medium (e.g., a magneto-optical disk), a compact disc read only memory (CD-ROM), a CD-R, a CD-R/W, and a semiconductor memory (e.g., a mask ROM, a programmable ROM (PROM), an erasable ROM (EPROM), a flash ROM, and a RAM). The program may be supplied to the computer by various types of transitory computer readable media. Examples of the transitory computer readable media include an electric signal, an optical signal, and an electromagnetic wave. The transitory computer readable media can supply the program to the computer via a wired communication channel such as an electric wire and an optical fiber or a wireless communication channel.

(63) The step of writing the program in the recording medium include processes performed chronologically in order, and also include processes that are not necessarily performed chronologically in order, but performed in parallel or individually.

(64) In other words, the disclosed control device and control method can be achieved in the form of various embodiments described below.

(65) (1) The robot control device **30** of the present disclosure is a control device that controls a robot **10** and a linear tool **20** that is attached to an end of an arm of the robot **10** and moves a workpiece in one direction, the control device comprising: an operation program analysis unit **311** configured to calculate a total travel range of the robot **10** and the linear tool **20** based on an operation program that sets an initial position of the linear tool **20** relative to a coordinate origin of the robot **10**; and a robot/tool travel distance calculation unit **312** configured to set a travel amount of the linear tool **20** that the linear tool **20** takes to reach a boundary position of the travel range to a specified value based on the calculated travel range, and calculate a travel amount of the end of the arm of the robot **10** when the linear tool **20** has traveled to the boundary position and travel amounts that the linear tool **20** and the end of the arm of the robot **10** would be required to take to reach at least one teaching point P_n other than the boundary position.

(66) With this robot control device **30**, an operator can create the operation program by teaching the single teaching point only, irrespective of the experience of the operator.

(67) (2) In the robot control device **30** of aspect (1), the specified value may be the maximum value of a distance that the linear tool **20** is able to travel.

(68) This allows the robot control device **30** to maximize the performance of the linear tool **20**.

(69) (3) The robot control device **30** of aspect (1) may further include an input unit **32** configured to receive an input from a user, and the specified value may be a value specified by the user via the input unit **32**.

(70) This allows the robot control device **30** to move the robot **10** and the linear tool **20** in any

selected travel range.

(71) (4) The robot control device **30** of any one of aspects (1) to (3) may further include an operation control/drive unit **313** configured to set a travel amount of the linear tool **20** and a travel amount of the end of the arm of the robot **10** to reach at least one teaching point other than the calculated boundary position in the operation program, and executes the operation program.

(72) This allows the robot control device **30** to drive the robot **10** and the linear tool **20**.

(73) (5) A control method of the present disclosure is a method of controlling a robot **10** and a linear tool **20** that is attached to an end of an arm of the robot **10** and moves a workpiece in one direction, the method comprising: an operation program analysis step of calculating a total travel range of the robot **10** and the linear tool **20** based on an operation program that sets an initial position of the linear tool **20** relative to a coordinate origin of the robot **10**; and a robot/tool travel distance calculation step of setting a travel amount of the linear tool **20** that the linear tool **20** would be required to take to reach a boundary position of the travel range to a specified value based on the calculated travel range, and calculating a travel amount of the end of the arm of the robot **10** when the linear tool **20** has traveled to the boundary position and travel amounts that the linear tool **20** and the end of the arm of the robot **10** would be required to take to reach at least one teaching point other than the boundary position.

(74) This control method can provide the same advantages as those of the device of aspect (1).

EXPLANATION OF REFERENCE NUMERALS

(75) **1** Robot system **10** Robot **20** Linear tool **21** Frame **22** Robot-side slider **23** Workpiece-side slider **30** Robot control device **31** Control unit **311** Operation program analysis unit **312** Robot/tool travel distance calculation unit **313** Operation control/drive unit **32** Input unit **33** Storage unit **34** Display unit

Claims

1. A control device for controlling a robot and an articulated shaft that is attached to an end of an arm of the robot and moves a workpiece in one direction, the control device comprising: an operation program analysis unit configured to calculate a travel range of the robot and the articulated shaft based on an operation program that sets an initial position of the articulated shaft relative to a coordinate origin of the robot; and a robot/tool travel distance calculation unit configured to set, to a specified value based on the calculated travel range, a travel amount that the articulated shaft would be required to take to reach a boundary position of the travel range, and calculate, from coordinates of the articulated shaft relative to the coordinate origin of the robot when the articulated shaft has traveled to the boundary position and from the specified value, a travel amount of the end of the arm of the robot when the articulated shaft has traveled to the boundary position and travel amounts that the articulated shaft and the end of the arm of the robot would be required to take to reach at least one teaching point other than the boundary position, wherein the robot and the articulated shaft are controlled by the control device based on the calculated travel amount of the end of the arm of the robot when the articulated shaft has traveled to the boundary position and the calculated travel amounts that the articulated shaft and the end of the arm of the robot would be required to take to reach at least one teaching point other than the boundary position.

2. The control device of claim 1, wherein the specified value is the maximum value of a distance that the articulated shaft is able to travel.

3. The control device of claim 1, further comprising: an input unit configured to receive an input from a user, wherein the specified value is a value specified by the user via the input unit.

4. The control device of claim 1, further comprising: an operation control/drive unit configured to set a travel amount of the articulated shaft and a travel amount of the end of the arm of the robot to reach at least one teaching point other than the calculated boundary position in the operation

program, and executes the operation program.

5. A method of controlling a robot and an articulated shaft that is attached to an end of an arm of the robot and moves a workpiece in one direction, the method comprising: an operation program analysis step of calculating a travel range of the robot and the articulated shaft based on an operation program that sets an initial position of the articulated shaft relative to a coordinate origin of the robot; and a robot/tool travel distance calculation step of setting, to a specified value based on the calculated travel range, a travel amount that the articulated shaft would be required to take to reach a boundary position of the travel range, and calculating, from coordinates of the articulated shaft relative to the coordinate origin of the robot when the articulated shaft has traveled to the boundary position and from the specified value, a travel amount of the end of the arm of the robot when the articulated shaft has traveled to the boundary position and travel amounts that the articulated shaft and the end of the arm of the robot that would be required to take to reach at least one teaching point other than the boundary position, wherein the robot and the articulated shaft are controlled by the control device based on the calculated travel amount of the end of the arm of the robot when the articulated shaft has traveled to the boundary position and the calculated travel amounts that the articulated shaft and the end of the arm of the robot would be required to take to reach at least one teaching point other than the boundary position.
